

### Profile Summary

Highly accomplished Ph.D. Researcher and Software Engineer with over 7 years of experience across industrial automation, Big Data, and advanced AI/ML. Skilled in real-time data streaming, formal verification, and full-stack development. Currently pursuing an industrial Ph.D. in collaboration with Technoleader SRL, combining academic rigor with practical product engineering. Proven track record of designing, optimizing, and deploying complex, high-throughput, and secure systems.

### Research and Current Engagement

- Nov 2023 – **Ph.D. Researcher**, *University of Genova & COBO Group*, Genova, Italy
- Present
- Conducted an in-depth literature review on **formal verification** and **runtime verification** of PLC systems.
  - Developing a **Runtime Verification (RV) framework** for Programmable Logic Controller (PLC) systems to enhance industrial safety and reliability.
  - Designed and implemented the framework using **C as a specification language**, enabling integration with existing PLC workflows.
  - Simulated and experimented with the proposed framework to obtain and analyze meaningful results.
  - Contributed to software development tasks and the design of key concepts for PLC Integrated Development Environments (IDEs).
  - Developed a **Structured Text (ST) to C transpiler** for improved cross-platform compatibility and performance.
  - Leveraged **AI tools** (Gemini, Perplexity) for research and **coding assistants** (VS Code Copilot, Cursor) to optimize the development workflow.
- Nov 2024 – **Teaching Support Activity**, *Distributed Computing Course*, University of Genova
- Feb 2025
- Provided academic and technical assistance to graduate students in the **Distributed Computing** course.

### Professional Experience

- Apr 2020 – **Freelance Software Engineer**, *Freelance*, Calicut, India, 1 year 1 month
- present
- Led the software development lifecycle for an educational math practice platform used by over 5,000 students and 200+ schools.
  - Designed and implemented a scalable architecture and database schema to support multi-user access using **Laravel 7, HTML/CSS, and MySQL**.
  - Managed client communication, gathered requirements, and coordinated UI integration, ensuring successful system deployment and post-launch maintenance.
- May 2018 – **Software Engineer**, *Bluecast Technologies*, Kochi, India, 1 year 10 months
- Mar 2020
- Project:** Security Information and Event Management (SIEM) Tool.
- Designed and optimized a real-time data streaming pipeline leveraging **Apache Spark, Kafka, HBase, and PostgreSQL**, handling over 30 million daily events for network analytics.
  - Engineered a high-throughput, optimized Rule Engine to process streaming data and enhance threat detection accuracy.
- Sep 2017 – **Freelance Web Developer**, *Freelance*, 7 Months
- Apr 2018
- Developed and maintained multiple static and dynamic websites, applying proficiency in **HTML, CSS, JavaScript, jQuery, and PHP**.

- Jan 2016 – **Desktop Support Engineer**, *Freelance*, Bangalore & Calicut, India, 1 year 6 months
- Jul 2017 ○ Provided direct technical support across multiple organizations, specializing in complex hardware/software troubleshooting and systems maintenance.
- Applied network and system administration knowledge to ensure secure and operational computing environments.

## Training and Certifications

- Apr 2020 – **Data Science Trainee**, *IPSR SOLUTIONS LIMITED*, Calicut, India, 5 Months
- Aug 2020 ○ Gained hands-on expertise in the ML lifecycle, including data collection, preprocessing (**Pandas**, **NLTK**), algorithm design (**Scikit-Learn**, **PyTorch**, **TensorFlow**), and actionable data visualization.
- Jul 2017 – **System Administration Training**, *Spectrum Softtech Solutions*, Kochi, India, 3 Months
- Sep 2017 ○ Mastered **Linux administration** and shell scripting for server configuration (Red Hat, NFS/Samba) and managed cloud deployments on **AWS** (EC2, S3).
- May 2017 **MCSA (Microsoft Certified Solutions Associate)**.
- Dec 2015 – **Certified IT Architect Course** (Focused on Desktop Engineering, CCNA, MCSA).
- Jun 2016

## Education

- Nov 2023 – **Ph.D. in Computer Science**, *University of Genova*, Genova, Italy
- Present
- Sep 2020 – **M.Sc. in Computer Science (Data Science & AI)**, *University of Genova*, Genova, Italy, GPA: 102/110
- Oct 2023
- Jul 2014 – **Master of Computer Applications (MCA)**, *Anna University*, Coimbatore, India, CGPA: 7.85/10
- Jun 2016
- Jun 2011 – **Bachelor of Computer Applications (BCA)**, *University of Calicut*, Malappuram, India, CGPA: 2.60/4
- Apr 2014

## Skills and Expertise

- Data Science & ML **Libraries:** Python (Pandas, NumPy), PyTorch, TensorFlow, Scikit-Learn, NLTK. **Techniques:** Deep Learning (GANs, CNNs, LSTMs), Formal Verification, Time-Series Analysis, Multi-Agent Systems.
- Software Engineering **Backend:** Java Spring, **Fast API**, PHP Laravel 7. **Frontend:** HTML, CSS, JavaScript, Next.js, **Pyside 6**. **Big Data:** Apache Spark, Kafka, HBase, Flume. **Databases/OS:** PostgreSQL, MySQL, Linux, AWS.
- AI & Productivity Tools **AI/Agents:** Gemini, ChatGPT, Perplexity, VS Code Copilot, Cursor. **Core:** Research, Transpiler Development, Problem Solving, Analytical Skills.
- Visualization & Languages **Tools:** Matplotlib, Seaborn, Plotly, Tableau, D3.js. **Languages:** Malayalam (Native), English (C1), Tamil (B1), Italian (A2).

## Academic Projects

- RMLGym **Runtime Verification for RL:** Developed RMLGym, a novel framework integrating **Runtime Verification (RV)** with Reinforcement Learning (RL) to generate formal property-based rewards.
- Speaker Identification **Speech Processing:** Engineered two robust speaker identification systems utilizing **MFCC+SVM** and **FFT+1D-CNN**, achieving up to 98% accuracy, and including a **Tkinter GUI** for practical application.

|                                  |                                                                                                                                                                                                                                                     |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| COVID-19 Dashboard               | <b>Data Visualization:</b> Developed a dynamic, interactive web-based dashboard using <b>D3.js</b> to visualize global COVID-19 spread and vaccination data, utilizing choropleth maps and custom charts for intuitive data exploration.            |
| Real vs. AI Image Classification | <b>Computer Vision Classification:</b> Developed a high-performance binary classifier (93.4% accuracy) using an optimized <b>AlexNet</b> CNN architecture to distinguish authentic from AI-generated images, including Grad-CAM for explainability. |
| Sigma2RML Transpiler             | <b>Transpiler Development:</b> Engineered a transpiler to convert Sigma rules into RML, enabling seamless integration with formal verification frameworks.                                                                                          |
| Text Summarization               | Engineered dual summarization pipelines: Extractive (ARTEX with Ultra-Stemming) and Abstractive (GAN with Seq2Seq encoder-decoder), assessing performance via ROUGE metrics.                                                                        |
| Timeseries Anonymization         | <b>Data Privacy:</b> Implemented and evaluated the (k,P)-anonymity privacy model using Naive and Kapra algorithms to protect time-series data from linkage attacks while preserving key patterns.                                                   |
| Image Generation                 | Constructed and trained Deep Convolutional GANs (DCGANs) for unsupervised image generation across complex datasets (CelebA, vehicles), showcasing expertise in generative modeling.                                                                 |
| Multi-Agent System               | <b>Decentralized Marketplace:</b> Designed a marketplace simulation using the <b>Jason framework</b> (BDI model), implementing autonomous agents for communication, coordination, and price negotiation.                                            |
| Fourier vs. Wavelet Transform    | <b>Signal Analysis:</b> Conducted a comparative study demonstrating the superiority of the Wavelet Transform for localizing features in time for non-stationary signals compared to the Fourier Transform.                                          |
| Mandelbrot Set Parallelization   | <b>High-Performance Computing:</b> Optimized the Mandelbrot Set generation algorithm by designing and implementing a parallelized version, significantly reducing execution time over sequential methods.                                           |

## Publications

- Peer-Reviewed Paper 1 Unniyankal H, Belardinelli F, Ferrando A, Malvone V. *"RMLGym: a Formal Reward Machine Framework for Reinforcement Learning."* CEUR Workshop Proceedings, Vol-3579/paper1.pdf. **URL:** <https://ceur-ws.org/Vol-3579/paper1.pdf>
- Peer-Reviewed Paper 2 Unniyankal H, Ancona D, Ferrando A, Parodi F, Alessi A, Bottino F. *"Runtime Verification of Program Organization Units in Safe Programmable Logic Controller Systems."* Conference on Dependable Systems and Networks (DSN) 2025. **DOI:** [10.1109/DSN-S65789.2025.00050](https://doi.org/10.1109/DSN-S65789.2025.00050)